

Phenological responses to climate change do not exhibit phylogenetic signal in a subalpine plant community

PAUL J. CARADONNA^{1,2,4} AND DAVID W. INOUE^{1,3}

¹Rocky Mountain Biological Laboratory, Crested Butte, Colorado 81224 USA

²Department of Ecology and Evolutionary Biology, University of Arizona, Tucson, Arizona 85701 USA

³Department of Biology, University of Maryland, College Park, Maryland 20742 USA

Abstract. Phylogenetic relationships may underlie species-specific phenological sensitivities to abiotic variation and may help to predict these responses to climate change. Although shared evolutionary history may mediate both phenology and phenological sensitivity to abiotic variation, few studies have explicitly investigated whether this is the case. We explore phylogenetic signal in flowering phenology and in phenological sensitivity to temperature and snowmelt using a 39-year record of flowering from the Colorado Rocky Mountains, USA that includes dates of first, peak, and last flowering, and flowering duration for 60 plant species in a subalpine plant community. Consistent with other studies, we found evidence in support of phylogenetic signal in first flowering date. However, the strength and significance of that signal were inconsistent across other measures of flowering in this plant community: peak flowering date exhibited the strongest phylogenetic signal, followed by first flowering date; last flowering date and duration of flowering exhibited patterns indistinguishable from random trait evolution. In contrast to first and peak flowering date, phenological sensitivities of all flowering measures to temperature and snowmelt did not exhibit a phylogenetic signal. These findings show that within ecological communities, phylogenetic signal in phenology does not necessarily imply phylogenetic signal in phenological sensitivities to abiotic variation.

Key words: *Blomberg's K; flowering time; Pagel's Lambda; phenological shifts; phenology; phylogenetic niche conservatism; Rocky Mountain Biological Laboratory; species specificity.*

INTRODUCTION

Shifts in phenology have emerged as one of the strongest bio-indicators of changing abiotic conditions across the globe (Sparks and Menzel 2002, Parmesan and Yohe 2003). Although there is an overwhelming trend for earlier phenological events in many ecosystems, considerable species-specific variation is evident in phenological responses to climate change (Fitter and Fitter 2002, Parmesan 2007, Crimmins et al. 2010, CaraDonna et al. 2014). If closely related taxa within ecological communities resemble each other in their phenological sensitivity to abiotic variation, then relatedness may help to predict species vulnerability, as well as community and ecosystem change, under future climate change scenarios (Willis et al. 2008, Cleland et al. 2012, Wolkovich et al. 2013a).

Investigation of phylogenetic relationships in plant phenology from a wide range of taxa from across the globe has revealed that phenological events in plants generally exhibit nonrandom phylogenetic patterns: that is, closely related taxa tend to resemble each other in the timing of their life-history events (e.g., Wright and Calderon 1995, Davies et al. 2013, Wolkovich et al.

2013b). Accordingly, the hypothesis has emerged that phenological sensitivity to changing abiotic conditions (i.e., climate-driven phenological shifts) likely exhibits phylogenetic signal as well. This hypothesis is based on the idea of conserved physiological responses to environmental cues among closely related taxa that mediate both phenology and phenological change (Davis et al. 2010, Davies et al. 2013). Despite this logical prediction, the little evidence available does not consistently support it. For example, Davis et al. (2010) reported that closely related taxa from two plant communities showed similar phenological sensitivities to temperature, but these same taxa did not show similar shifts in first flowering date through time. Similarly, Wolkovich et al. (2013b) found no evidence of phylogenetic signal for shifts in first flowering for taxa from several different plant communities in North America.

Furthermore, no studies to date have investigated whether phylogenetic patterns exhibited in mean dates of first flowering are consistent with other measures of phenology, in part due to a paucity of long-term records for which multiple phenological measures are available for the majority of taxa in a community. First, peak, and last flowering, and duration of flowering have been shown to shift independently of one another under climate change (CaraDonna et al. 2014), leaving open the potential for phylogenetic signal to vary across these component measures.

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⁴ E-mail: pcaradonna@gmail.com

Herein we explore phylogenetic patterns in plant phenology and phenological sensitivity to abiotic factors using a 39-year flowering phenology data set from the Colorado Rocky Mountains, USA that includes information on dates of first, peak, and last flowering, as well as flowering duration. We focus our investigation on 60 representative taxa from this subalpine plant community for which we have sufficient data to assess phenological change accurately. Specifically, we test for phylogenetic signal in flowering time and in phenological responses to climate change. We define flowering time as mean flowering date across years, and we define phenological responses to climate change as shifts in flowering phenology through time, and sensitivity of flowering time to snowmelt date and temperature. We ask two primary questions: (1) are phylogenetic patterns similar between phenology and phenological change within this community, and (2) are phylogenetic patterns consistent among the different component measures of a phenology distribution (first, peak, last, and duration)? Based on previous findings (Davies et al. 2013), we expected flowering time to exhibit nonrandom phylogenetic patterns, in part due to conserved evolutionary responses associated with the appropriate abiotic conditions for optimal timing of reproduction. However, because of the considerable variation in phenological responses to climate change among species in this community (Iler et al. 2013a, CaraDonna et al. 2014), we did not expect closely related taxa to exhibit similar phenological sensitivity to abiotic variation. For this same reason, we expected to observe differences in phylogenetic patterns among first, peak, last, and duration of flowering.

MATERIALS AND METHODS

Flowering phenology data

Flowering phenology data were collected at the Rocky Mountain Biological Laboratory (RMBL) in Gothic, Colorado, USA (38°57.5' N, 106°59.3' W, 2900 m above sea level). The subalpine landscape surrounding the RMBL is a mosaic of wet and dry meadows, and aspen and spruce forest. For each flowering plant species that occurs in our 30 2 × 2 m plots, either the number of open flowers per stalk or the number of flowering inflorescences (for species with many small flowers) were counted every other day throughout the entire growing season from 1974 to 2012. This allowed us to examine multiple components of a phenological distribution: first, peak, and last day of flowering, as well as flowering duration. Although 121 species occur in these permanent plots, we limit our analyses of phylogenetic signal in flowering traits to the 60 species for which we have sufficient data to assess phenological shifts accurately. This sampling of 60 species includes 54 genera representing 80% (24 out of 30) of the vascular plant families present in this plant community (see Appendix A: Table A1 for details). For additional details about the data set, study site, and quantification of phenological change,

see Inouye et al. (2002), Inouye (2008), Iler et al. (2013a), and CaraDonna et al. (2014).

Climate data

Over the course of our study period (1974–2012), summer air temperatures have increased by $0.4^{\circ} \pm 0.1^{\circ}\text{C}$ (mean $\pm$ SE) per decade, and the date of spring snowmelt has advanced by 3.5 ± 2.0 days per decade (Iler et al. 2013a). These changes in abiotic conditions are consistent with global patterns of climate change that have occurred since the 1970s (IPCC 2007). We use both mean summer temperature and the date of snowmelt in our analyses because they represent two biologically independent abiotic factors important for flowering phenology in this ecosystem. Mean summer temperatures used in analyses (June–August) are the average of the daily minimum and maximum temperatures at the Crested Butte National Oceanic and Atmosphere Administration weather station, Colorado (~9 km south of the RMBL). Furthermore, June, July, and August temperatures are all positively correlated with one another; thus, they represent a composite measure of summer temperature. Records for the annual timing of snowmelt in this ecosystem come from a permanent 5 × 5 m plot at the RMBL, in which the first day of bare ground is recorded as the date of snowmelt (snowpack measurements are recorded daily).

Quantifying flowering phenology and phenological responses to climate change

We quantified mean date of flowering for first, peak, last, and duration of flowering in two ways: (1) averaged across the first 10 years of the data set (1974–1984), and (2) averaged across all 39 years of the data set (1974–2012). Averaging flowering time values across all 39 years, a time period when phenological shifts are known to have occurred (CaraDonna et al. 2014), introduces more error into our measures of flowering time for each taxon, which can influence estimates of phylogenetic signal (Blomberg et al. 2003). Furthermore, flowering time values averaged across the first 10 years of the data set more likely represent a pre-climate-change baseline (sensu Iler et al. 2013b). The date of first flowering reported here is the first day on which a species is observed in flower in any of our permanent plots; peak flowering was calculated as the day on which 50% of flowers of a given species were recorded as open; last flowering is the last day on which a species was observed in flower in any plot; and duration of flowering is the number of days a species was recorded as in flower (CaraDonna et al. 2014).

We quantified phenological responses of flowering time to climate change in three ways: (1) shifts in flowering phenology through time (1974–2012), (2) shifts in flowering phenology in response to snowmelt date (snowmelt sensitivity), and (3) shifts in flowering phenology in response to summer temperature (temperature sensitivity). The timing of snowmelt date and

temperature are strongly associated with shifts in flowering phenology in this study system (Inouye et al. 2002), independently explaining on average 68% and 66% of the interannual variation in flowering phenology across species, respectively (Iler et al. 2013a). Compared to phenological sensitivities to snowmelt date and summer temperature, shifts in flowering phenology through time provide the most conservative measure of phenological change and the measure with the highest variation. We include phenological shifts through time in our analyses because such shifts are a widely used metric of phenological change, especially when information on local abiotic factors is unavailable or unreliable. Trait values for shifts in flowering phenology through time for each plant species were quantified as the slope of a linear regression model of flowering date vs. year; trait values for phenological sensitivity to snowmelt were quantified as the slope of a linear regression model of flowering date vs. snowmelt date; and trait values for phenological sensitivity to temperature were quantified as the slope of a linear regression model of flowering date vs. June-to-August mean temperature. All flowering time data used in analyses are reported in Appendix A: Table A1 and Appendix B: Table B1, and are available elsewhere, i.e., Iler et al. (2013a) and CaraDonna et al. (2014).

Testing for phylogenetic signal

To conduct phylogenetic analyses, we used a pruned molecular phylogeny that is fully resolved down to genus for 1246 taxa and includes the vast majority of the taxa in this study (see Davies et al. [2013] for the complete phylogeny and the Supplement for our community phylogeny). The two taxa from our community that were not included in the molecular phylogeny were placed within the nearest clade. We tested for phylogenetic signal in flowering phenology and phenological responses to climate change using two commonly employed metrics: Blomberg's K and Pagel's λ (Pagel 1999, Blomberg et al. 2003). Because Blomberg's K and Pagel's λ differ in the way they test for phylogenetic signal, which can generate some degree of inconsistency in results, we use both metrics to ensure appropriate interpretation of patterns in our plant community (Münkemüller et al. 2012). Our intention was not to explicitly compare inconsistencies between methods, but instead, to reveal cases in which we obtained inconsistent results, and to treat them with caution. To test the robustness of our results to variation in phylogenetic branch length and topology, we compared our results from the molecular genus tree (Table 1; Supplement) to those generated from a less well-resolved phylogeny (Wikström et al. 2001, following Davies et al. 2013); the two phylogenies differ in branch lengths and topology above the genus level (see Appendix D: Table D1 for details).

Both Pagel's λ and Blomberg's K compare the observed distribution of trait values with the analytical

expectation based on the topology of the phylogeny and assume trait evolution under Brownian motion (Pagel 1999, Blomberg et al. 2003, Ackerly 2009). K values that are approximately equal to 1 match trait evolution under the Brownian motion expectation, and indicate phylogenetic signal; K values much less than 1 and closer to 0 indicate little or no phylogenetic signal associated with random trait evolution or convergence; K values greater than 1 indicate strong phylogenetic signal and conservatism of traits. We evaluated the statistical significance of phylogenetic signal in Blomberg's K by comparing the observed patterns of phylogenetic independent contrasts of each trait value relative to a null model of randomly shuffling trait values across the phylogeny, as implemented in the R package Picante (Kembel et al. 2010, R Development Core Team 2014). In contrast, Pagel's λ is a scaling parameter that ranges from 0 to 1. Lambda values of 0 indicate no phylogenetic signal, whereas values of 1 indicate perfect phylogenetic signal, matching trait evolution under Brownian motion (Pagel 1999). We estimated the statistical significance of phylogenetic signal using Pagel's λ by comparing our observed trait values to those expected on a phylogeny with λ values of 0 (i.e., no phylogenetic signal); thus statistical significance indicates that trait values are nonrandomly distributed across the phylogeny. Calculations of Pagel's λ and significance tests were conducted using the R package Geiger (Harmon et al. 2008, R Development Core Team 2014).

RESULTS

Overall, our tests of phylogenetic signal using Blomberg's K and Pagel's λ were consistent: in only one case did we detect significant phylogenetic signal using one metric and not with the other (first flowering time; Table 1). Similarly, phylogenetic signal was quantitatively and qualitatively consistent using either mean flowering time values across the first 10 years of the data set (1974–1984) or mean values across the entire 39 years of the data set (1974–2012; Table 1 and Appendix C: Table C1); additionally, phylogenetic signal in flowering time (peak) did not vary directionally across the 39-year study period ($R^2 = 0.037$, $P = 0.253$). We therefore focus further discussion of flowering time on the mean date of flowering calculated over the first 10 years of the data set because less error is associated with this measure, and because it more likely represents a pre-climate change baseline compared to the 39-year mean over which phenological shifts are known to have occurred (CaraDonna et al. 2014).

The strength and significance of phylogenetic signal in flowering phenology varied across each of our different measures of flowering time (first, peak, last, and duration of flowering; Table 1). Peak flowering exhibited the strongest phylogenetic signal using either Blomberg's K or Pagel's λ (Fig. 1); first flowering exhibited a similarly strong nonrandom phylogenetic pattern, but

TABLE 1. Phylogenetic signal of flowering phenology traits for 60 species from a subalpine plant community in the Colorado Rocky Mountains, USA.

Phenological trait	Blomberg's <i>K</i>		Pagel's λ	
	Value	<i>P</i>	Value	<i>P</i>
Flowering time				
First flowering	0.683	0.041	0.434	0.120
Peak flowering	0.748	0.016	0.587	0.041
Last flowering	0.498	0.514	0.011	0.952
Duration of flowering	0.449	0.692	<0.001	0.999
Shift through time				
First flowering	0.342	0.907	<0.001	0.999
Peak flowering	0.351	0.894	<0.001	0.999
Last flowering	0.467	0.482	<0.001	0.999
Duration of flowering	0.445	0.563	<0.001	0.999
Snowmelt sensitivity				
First flowering	0.512	0.370	0.147	0.597
Peak flowering	0.413	0.701	<0.001	0.999
Last flowering	0.511	0.366	<0.001	0.999
Duration of flowering	0.633	0.060	<0.001	0.999
Temperature sensitivity				
First flowering	0.480	0.490	<0.001	0.999
Peak flowering	0.441	0.591	<0.001	0.999
Last flowering	0.485	0.393	<0.001	0.999
Duration of flowering	0.569	0.157	0.279	0.565

Notes: *K* significant from random at $P < 0.05$ and λ significant from zero at $P < 0.05$ are shown in bold. Flowering times shown here are calculated from the mean date of flowering for each species over the first 10 years of the data set (1974–1984); results for flowering time across all 39 years (1974–2012) are quantitatively similar and shown in Appendix C: Table C1. Values for shifts in flowering phenology through time, and phenological sensitivities to snowmelt date and temperature, are based on slopes of linear regression models from Caradonna et al. (2014); all values used in analyses are reported in Appendix A: Table A1 and Appendix B: Table B1.

only when calculated with Blomberg's *K* (values for Pagel's λ were qualitatively similar but marginally significant; Table 1). Last flowering and flowering duration exhibited no phylogenetic signal (Table 1). In contrast to first and peak flowering date, shifts in flowering phenology through time, phenological sensitivity to snowmelt date, and phenological sensitivity to temperature did not show phylogenetic signal for any measure of phenology (Table 1, Fig. 1b–d). Both Blomberg's *K* and Pagel's λ values were low, and in all cases were not significantly different from a random pattern of trait evolution (Table 1). In general, our estimates of phylogenetic signal were robust to variation in branch length and tree topology (Appendix D: Table D1).

DISCUSSION

Although phylogenetic signal in phenological responses to climate change may be present in some communities, even among those that are geographically distinct (Davis et al. 2010), our results provide evidence that this is not necessarily a ubiquitous phenomenon present in all plant communities. Shifts in flowering phenology through time and phenological sensitivities to snowmelt

date and temperature did not exhibit phylogenetic signal, indicating that more closely related taxa in this subalpine plant community do not share similar phenological sensitivities to abiotic variation (Table 1, Fig. 1). However, consistent with other studies (e.g., Davies et al. 2013, Wolkovich et al. 2013b), our results reveal that more closely related taxa do tend to commence flowering at more similar times during the growing season in this community than would be expected by a random pattern of trait evolution. It is important to note that our results pertain to phylogenetic patterns within an ecological community, rather than for species within a well-sampled clade; a community phylogeny inevitably represents a subsample of taxa from multiple clades, the results of which can provide insight into ecological patterns (e.g., species vulnerability, community change) rather than mechanisms underlying trait evolution.

The strength and significance of evidence for phylogenetic signal varied across different measures of flowering phenology. Mean date of peak flowering showed the strongest and most consistent pattern of phylogenetic signal (Table 1). Peak flowering may be the most robust measure of phylogenetic signal in flowering time because peak abundance represents an aggregate population response (Miller-Rushing et al. 2008). Although mean date of first flowering showed qualitatively similar patterns to peak flowering, only Blomberg's *K* detected significant phylogenetic signal in first flowering (Table 1). This discrepancy between first and peak flowering using Pagel's λ is likely attributed to two factors. First, unlike peak flowering, the date of first flowering is often represented by only a few individuals instead of representing an aggregate property of the population; this can introduce more variability into this measure of flowering time (sensu Miller-Rushing et al. 2008), which can decrease the likelihood of detecting phylogenetic signal (Blomberg et al. 2003). Second, simulation studies have shown that Pagel's λ is a more conservative measure of phylogenetic signal when compared to Blomberg's *K* (Münkemüller et al. 2012).

In contrast to first and peak flowering, we found no evidence for phylogenetic signal in mean date of last flowering or mean duration of flowering (Table 1). Unlike first and peak flowering, last flowering and flowering duration are affected by a suite of abiotic and biotic factors occurring after flowering onset, which can affect floral longevity (Primack 1985). For example, while warmer temperatures may advance the onset of flowering, resource availability (e.g., water and nutrients) can alter floral production, delay the end of flowering, and extend overall flowering duration (Campbell and Halama 1993). Similarly, biotic factors such as pollination may extend or contract the end of the flowering period (Primack 1985, Bolmgren et al. 2003). Furthermore, similar to first flowering date, last flowering, and by logical extension, flowering duration, is also often represented by a few individuals, rather

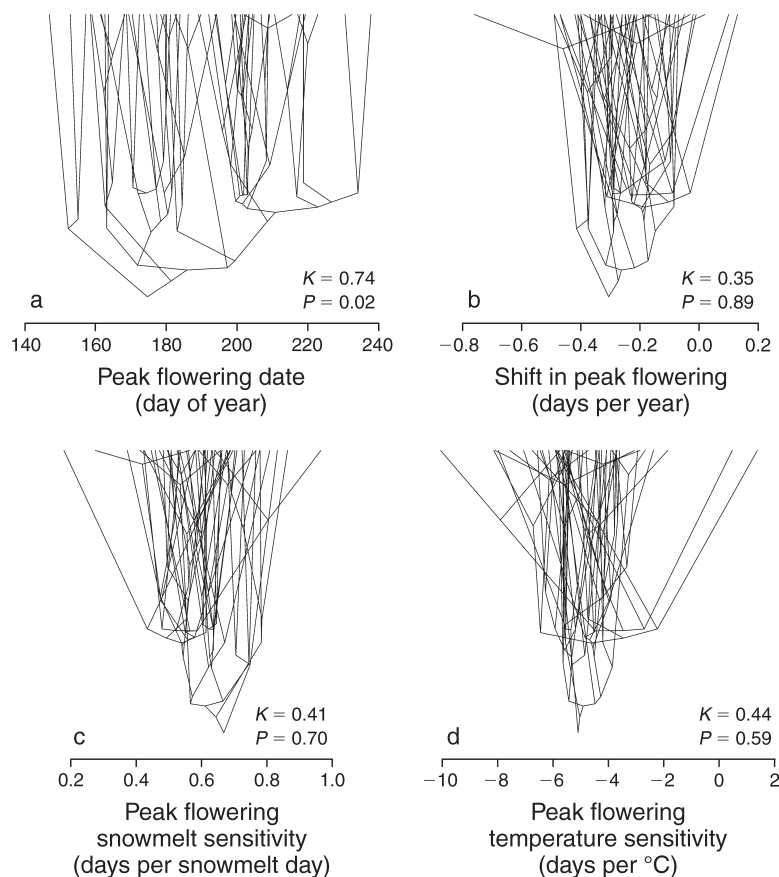


FIG. 1. Traitgrams illustrating phylogenetic signal in (a) mean date of peak flowering, (b) shift in peak flowering date through time, (c) shift in peak flowering in response to snowmelt date, and (d) shift in peak flowering in response to temperature. Blomberg's K and P values are shown for each signal. Tips on the tree are arranged by trait values; more crossover among branches within the tree indicates less phylogenetic signal. Mean date of peak flowering shown here is calculated from the first 10 years of the study period (1974–1984). This figure is presented for illustrative purposes and is not intended for interpretation of ancestral states.

than an aggregate property of the population. These results suggest that different component measures of a phenological distribution do not necessarily exhibit identical phylogenetic patterns. Therefore, measures of phenology that best capture aggregate population-level patterns (e.g., peak flowering) may provide the most reliable signal of phylogenetic patterns in phenological traits, compared to measures that represent only the early and late tail-ends of a population-level response (e.g., first and last flowering).

For all measures of changes in flowering phenology, more closely related taxa did not resemble each other in their phenological responses to climate change in this subalpine plant community (Table 1, Fig. 1b–d). Measurement error in both rates of phenological change and sensitivity to abiotic factors can influence the ability to detect phylogenetic signal. However, this is unlikely here because approximately 70% of the variation in flowering phenology in this ecosystem is independently explained by the timing of snowmelt and summer temperatures (Iler et al. 2013a). Furthermore, in all cases, tests of phylogenetic signal in phenological change

using either Blomberg's K or Pagel's λ revealed consistently low values that were indistinguishable from random patterns of trait evolution (Table 1, Fig. 1b–d). The only near exception was the sensitivity of flowering duration to snowmelt, which was marginally significant (using Blomberg's K ; Table 1). Snowmelt date is correlated with snowpack and hence the amount of water entering this ecosystem in the spring (Inouye et al. 2002). The ability of a species to flower for more or less time during the growing season may be influenced by floral longevity and other physiological mechanisms related to drought stress (Primack 1985), which may be relatively conserved among some taxonomic groups.

The hypothesis that conserved evolutionary responses to abiotic conditions mediate both phenology and phenological responses to local abiotic variation is logical; however, the evidence presented in this study does not support it. Our results provide a clear counterpoint to the expectation that phenological responses to climate change show strong phylogenetic patterns within plant communities. We propose two non-mutually exclusive explanations that may underlie

the discrepancy between phylogenetic signal in phenology vs. the absence of phylogenetic signal in phenological sensitivity to abiotic factors.

First, the presence of phylogenetic signal in flowering time does not imply strong phylogenetic conservatism (i.e., phylogenetic niche conservatism; Losos 2008), nor does it preclude closely related taxa from occupying different abiotic environments via differences in seasonality or habitat even if they share similar physiological pathways related to phenology. Differences in seasonality or habitat can act as environmental filters by which a phenological response can be mediated. In this subalpine ecosystem, the growing season is brief and changes rapidly within a year, thus slight differences in flowering time can represent considerably different abiotic environments, which have been shown to contribute to differences in phenological change among closely related taxa in this plant community (Miller-Rushing and Inouye 2009, Iler et al. 2013a, CaraDonna et al. 2014). Similarly, closely related taxa flowering at very similar times can occupy microhabitats that represent different abiotic environments, which can also mediate phenological shifts. For example, *Potentilla pulcherrima* and *P. hippiana* (Rosaceae) flower at virtually the same time during midsummer (peak flowering: 22 July $\pm$ 11 days and 23 July $\pm$ 10 days, respectively), but occupy wet meadows and dry rocky meadows, respectively, and do not exhibit similar phenological shifts (Iler et al. 2013a, CaraDonna et al. 2014).

Second, because phenology is highly heritable and closely tied to survival and reproduction, it plays a major role in determining the geographic ranges of plants (Rathcke and Lacey 1985, Chuine and Beaubien 2001, Etterson 2004, Franks et al. 2007, Forrest and Miller-Rushing 2010). Phenological sensitivity to local abiotic variation is likely shaped by interactions between local and regional climate conditions experienced by a species across its range. The strength of local vs. regional effects will in part depend on the balance between local adaptation and gene flow among populations. Assuming that all of the closely related, co-occurring species in a community do not have identically overlapping ranges, their phenological sensitivities likely differ, as do the climate conditions experienced across their ranges; thus, closely related taxa may exhibit different phenological sensitivities to local abiotic variation, despite similar physiological mechanisms that may underlie flowering time.

In sum, we hypothesize that local and regional climate conditions, combined with processes occurring during community assembly, will mediate phylogenetic patterns of both phenology and phenological responses to climate change within ecological communities. In contrast to previous work, the findings presented in this study show that within a plant community, phylogenetic signal in phenology does not necessarily imply phylogenetic signal in phenological sensitivities to abiotic variation. Assuming that phenology represents pheno-

logical change could lead to potentially inaccurate conclusions about the effects of shared evolutionary history on phenological traits and their responses to future abiotic change.

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SUPPLEMENTAL MATERIAL

Ecological Archives

Appendices A–D and the Supplement are available online: <http://dx.doi.org/10.1890/14-1536.1.sm>